
DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Experiment No.: 7

Interface LM 35 temperature sensor with LPC 2148.

Name: _____

Roll No.: _____ *Batch:* _____

Date of Performance: _____

Date of Assessment: _____

Particulars	Marks
Marks for Regularity (05)	
Marks for Performance (15)	
Understanding (Viva) (05)	
Total (25)	
Signature of Course Teacher	

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Experiment: 7

AIM: Interfacing LM35 temperature sensor with LPC2148.

OBJECTIVE:

1. To interface the LM35 precision temperature sensor with the LPC2148 ARM7 microcontroller and read real-time ambient temperature using the on-chip 10-bit ADC.
2. To understand working principle of the LM35 - a linear voltage output sensor that produces 10mV per °C – and relate the analog output voltage to the corresponding temperature in degree Celsius.

APPARATUS:

1. LPC2148 module development board.
2. LM 35 Temperature Sensor.
3. Connecting wire.
4. Adapter.

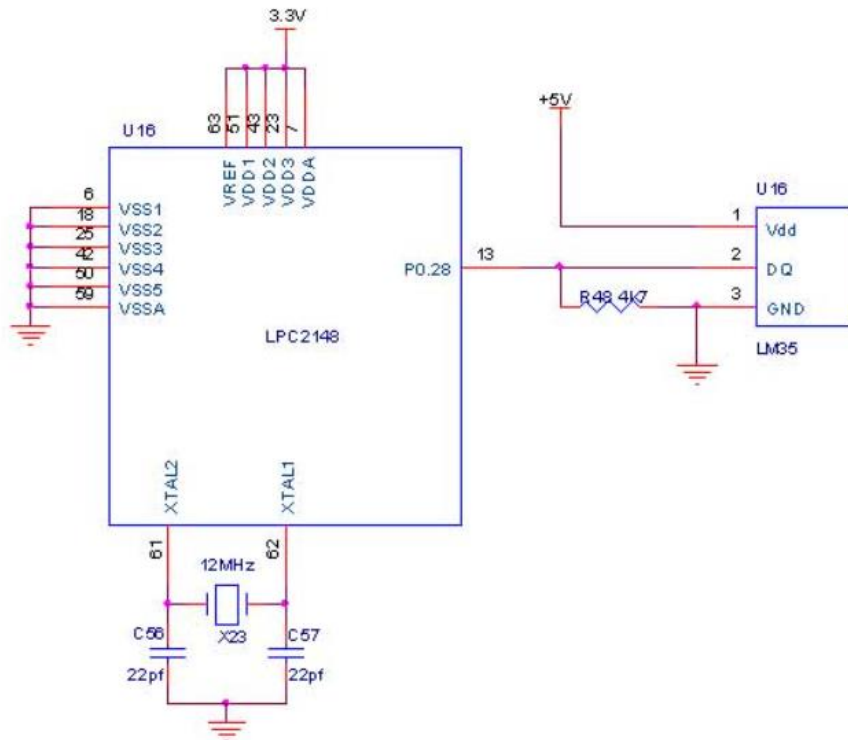
THEORY:

The LM35 is a precision integrated-circuit temperature sensor from Texas Instruments whose output voltage is linearly proportional to the Celsius temperature. It requires no external calibration.

Applications of LM 35 Temperature Sensor:

1. Environmental Monitoring: Used to measure ambient room temperature in homes, offices or weather stations.
2. HVAC Systems: Employed in heating, ventilation, and air conditioning systems for precise temperature regulation.
3. Battery Protection & Management: Used to monitor battery temperature to prevent overheating during charging or operation.
4. Industrial Control & Automation: Suitable for monitoring temperatures on production lines, such as in automobile manufacturing.
5. Thermal Management: Integrated into circuits to act as a temperature-sensitive switch or safety shutdown mechanism to prevent component damage.

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING



PIN DESCRIPTION:

Temperature sensor is connected to Port 0 at Pin 28 of LPC2148.

CONCLUSION:
